p8106_hw2_sc5826

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Data Cleaning & Splitting into Train/Test

```
college = read_csv("College.csv") |>
  janitor::clean_names()
```

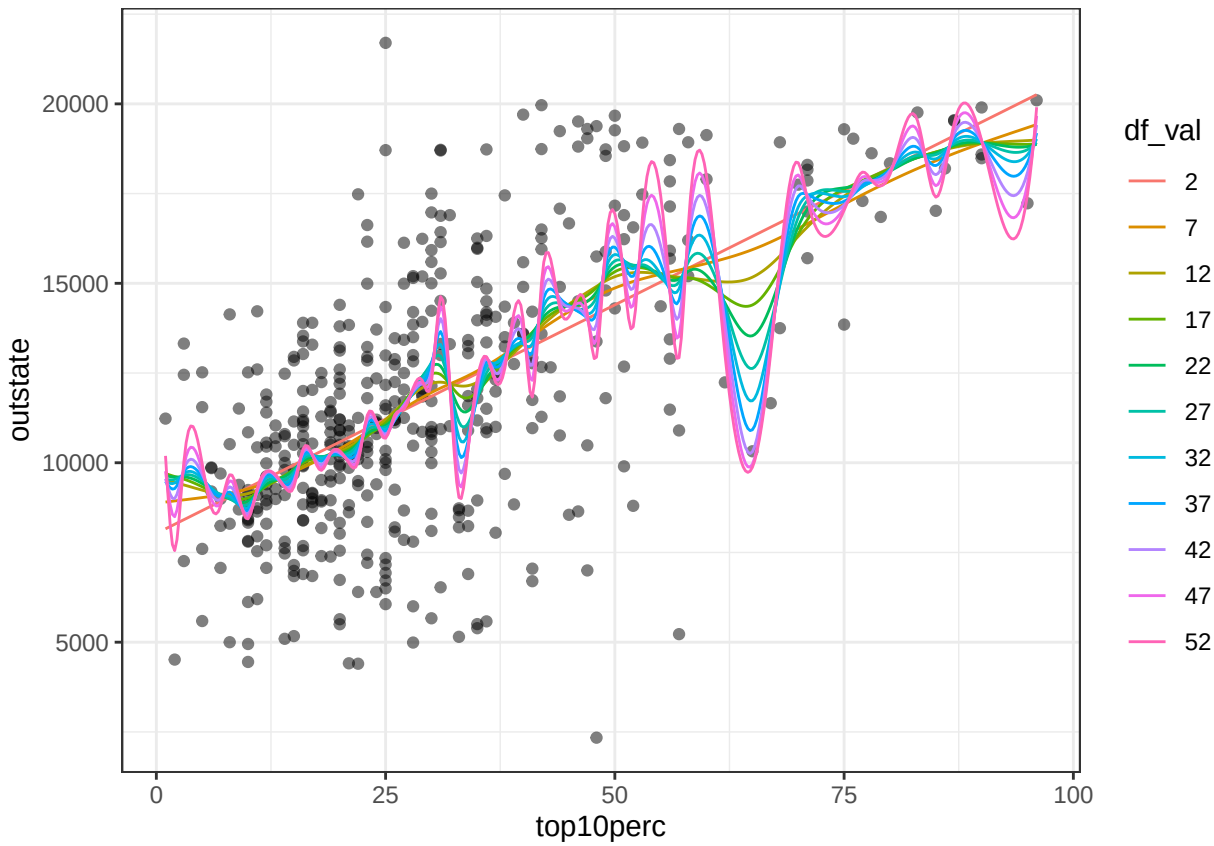
```
## Rows: 565 Columns: 18
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (1): College
## dbl (17): Apps, Accept, Enroll, Top10perc, Top25perc, F.Undergrad, P.Undergr...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
set.seed(1234)
indexTrain = createDataPartition(y = college$outstate, p = 0.8, list = FALSE)
college_train = college[indexTrain, ]
college_test = college[-indexTrain, ]
```

A - Smoothing Spline

```
set.seed(1234)
top10perc_grid = seq(min(college_train$top10perc), max(college_train$top10perc),
                     by = 0.25)

spline_df_fits =
  tibble(df_val = seq(from = 2, to = 52, by = 5)) |>
  mutate(
    spline_model = map(df_val, \(df) smooth.spline(x = college_train$top10perc,
                                                   y = college_train$outstate,
                                                   df = df)),
    predictions = map(spline_model, \(model) predict(model,
                                                     x = top10perc_grid) |>
                        as.data.frame())
  ) |>
  unnest(predictions) |>
  mutate(df_val = as.factor(df_val))

ggplot(college_train, aes(x = top10perc, y = outstate)) +
  geom_point(alpha = 0.5) +
  geom_line(data = spline_df_fits, aes(x = x, y = y, color = df_val)) +
```

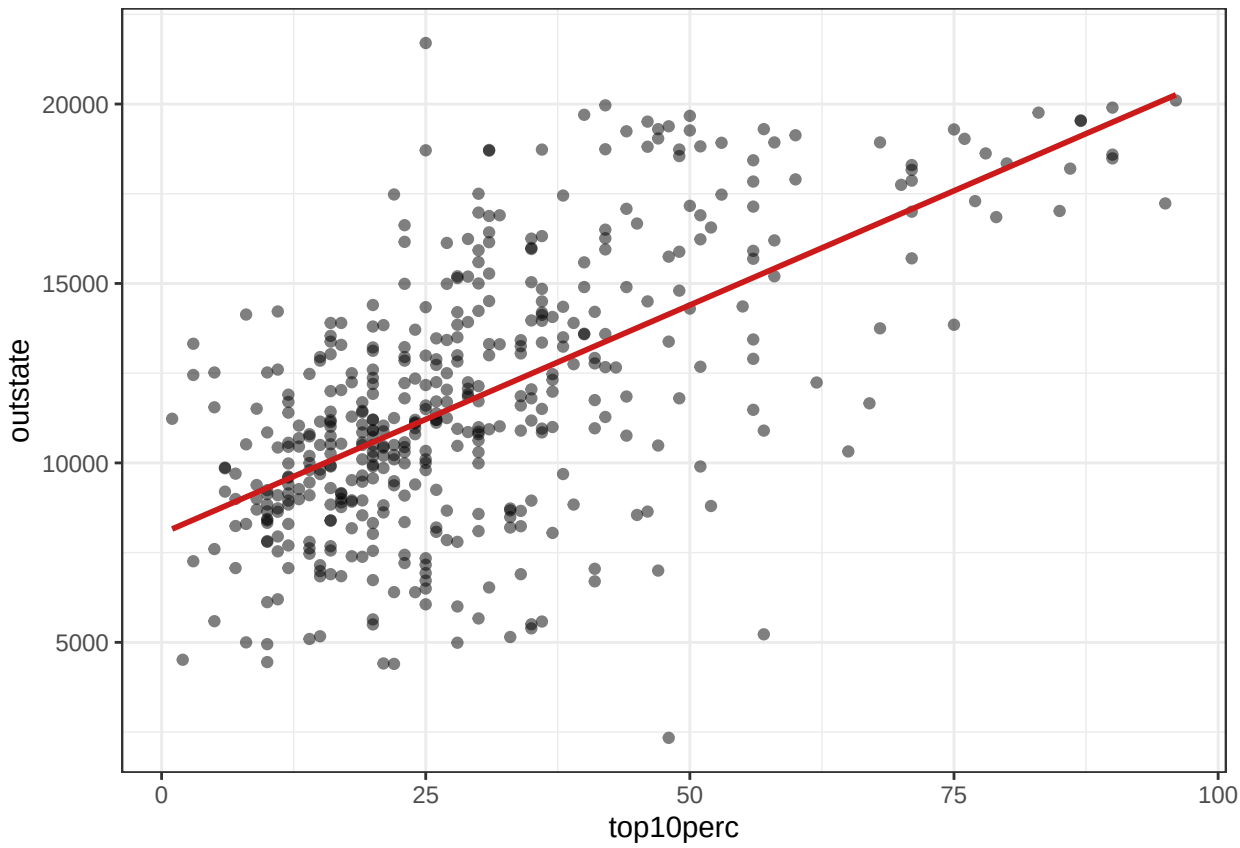
```
  theme_bw()
```



As the degrees of freedom increase, the smoothing spline line deviates more from a straight line. At the lowest degrees of freedom (`df = 2`), the model has a heavier penalty for roughness, resulting in a nearly straight line. At the other end (`df = 52`), there is a lighter penalty for roughness, and the spline adjusts more to individual points. At higher degrees of freedom, there is a risk of overfitting the training data.

To select the optimal degrees of freedom, I used the default smoothing spline function, without specifying the degrees of freedome. This calculates and minimizes the default Generalized Cross-Validation (GCV) error to select the tuning parameter (λ).

```
best_smoothing_spline = smooth.spline(x = college_train$top10perc,
                                      y = college_train$outstate)

optimal_df = best_smoothing_spline$df

optimal_pred_df = predict(best_smoothing_spline, x = top10perc_grid) |>
  as.data.frame()

ggplot(college_train, aes(x = top10perc, y = outstate)) +
  geom_point(alpha = 0.5) +
  geom_line(data = optimal_pred_df, aes(x = x, y = y),
            color = rgb(.8, .1, .1, 1), linewidth = 1) +
  theme_bw()
```

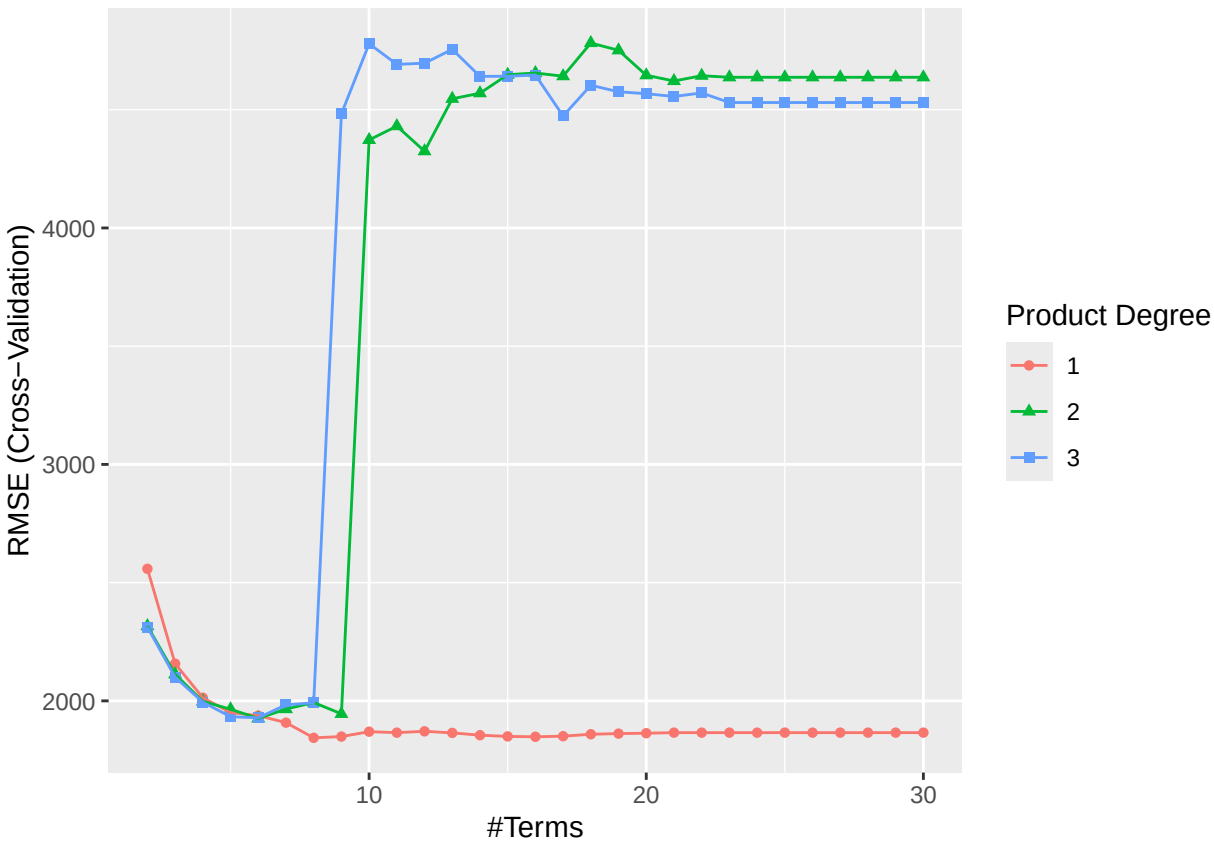


Based on the above criterion, the optimal degrees of freedom chosen for the smoothing spline model was ~2.

B - MARS

Train a multivariate adaptive regression spline (MARS) model to predict the response variable. Report the regression function. Present the partial dependence plot of an arbitrary predictor in your model. Report the test error.

```
# matrix of predictors
x = model.matrix(outstate ~ . - college, college_train)[, -1]
# vector of response
y = college_train$outstate

mars_grid = expand.grid(degree = 1:3, nprune = 2:30)

set.seed(1234)

ctrl1 = trainControl(method = "cv",
                     number = 10)

mars_model = train(x, y,
                   method = "earth",
                   tuneGrid = mars_grid,
                   trControl = ctrl1)

ggplot(mars_model)
```




```r
mars_model$bestTune
```

```
##   nprune degree
## 7      8      1
```

Regression Function:

```r
summary(mars_model$finalModel)
```

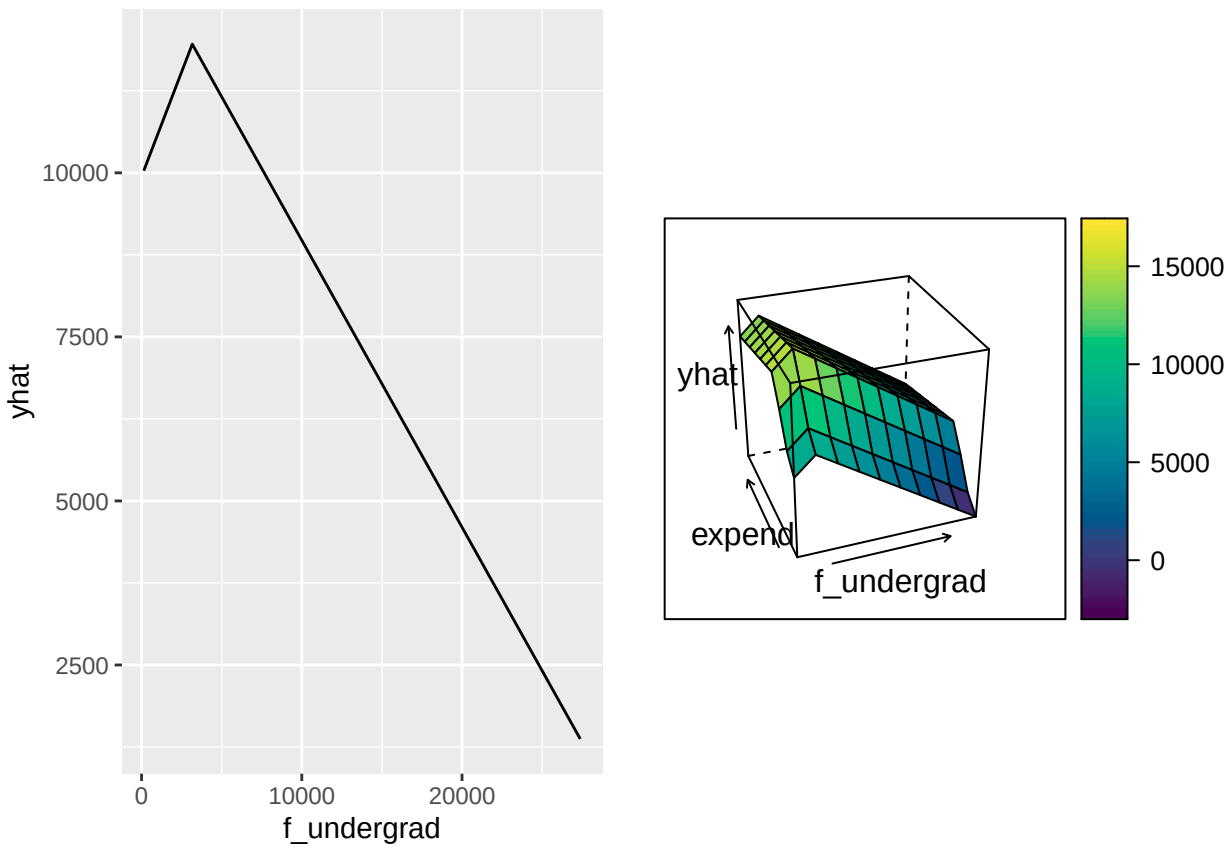
```
## Call: earth(x=matrix[453,16], y=c(7440,11250,75...), keepxy=TRUE, degree=1,
##             nprune=8)
##
##                     coefficients
## (Intercept)            9221.1968
## h(apps-2306)              0.4164
## h(1379-f_undergrad)      -2.1852
## h(f_undergrad-1379)      -0.4373
## h(4250-room_board)       -1.9481
## h(perc_alumni-14)        51.9742
## h(expend-5391)            0.7316
## h(expend-15494)          -0.8081
##
## Selected 8 of 28 terms, and 5 of 16 predictors (nprune=8)
## Termination condition: Reached nk 33
## Importance: expend, room_board, f_undergrad, perc_alumni, apps, ...
## Number of terms at each degree of interaction: 1 7 (additive model)
## GCV 3232127    RSS 1368795004    GRSq 0.765237    RSq 0.7795547
```

Final Regression Function:

$$
\begin{aligned}
\widehat{Outstate} = 9221.1968 \\
&+ 0.4164 * \text{h(apps-2306)} \\
&- 2.1852 * \text{h(1379-f.undergrad)} \\
&- 0.4373 * \text{h(f.undergrad-1379)} \\
&- 1.9481 * \text{h(4250-room.board)} \\
&+ 51.9742 * \text{h(perc.alumni-14)} \\
&+ 0.7316 * \text{h(expend-5391)} \\
&- 0.8081 * \text{h(expend-15494)}
\end{aligned}
$$

Partial Dependence Plot:

```
pdp_plot_1 = pdp::partial(mars_model, pred.var = c("f_undergrad"),
                          grid.resolution = 10) |>
  autoplot()

pdp_plot_2 = pdp::partial(mars_model, pred.var = c("f_undergrad", "expend"),
                          grid.resolution = 10) |>
  pdp::plotPartial(levelplot = FALSE, zlab = "yhat", drape = TRUE,
                   screen = list(z = 20, x = -60))

gridExtra::grid.arrange(pdp_plot_1, pdp_plot_2, ncol = 2)
```



Test Error:

```
predictions = predict(mars_model, newdata = college_test)
```

```r
test_rmse = RMSE(predictions, college_test$outstate)
```

The test error (RMSE) for the MARS model is $1861.76.

C - GAM

To start, I will be using the variables selected previously in the MARS model in the GAM model. I will further test for any meaningful interactions between `apps` (Number of applications received) and `f_undergrad` (Number of full-time undergraduates), as well as `room_board` (Room and board costs) and `expend` (Instructional expenditure per student).

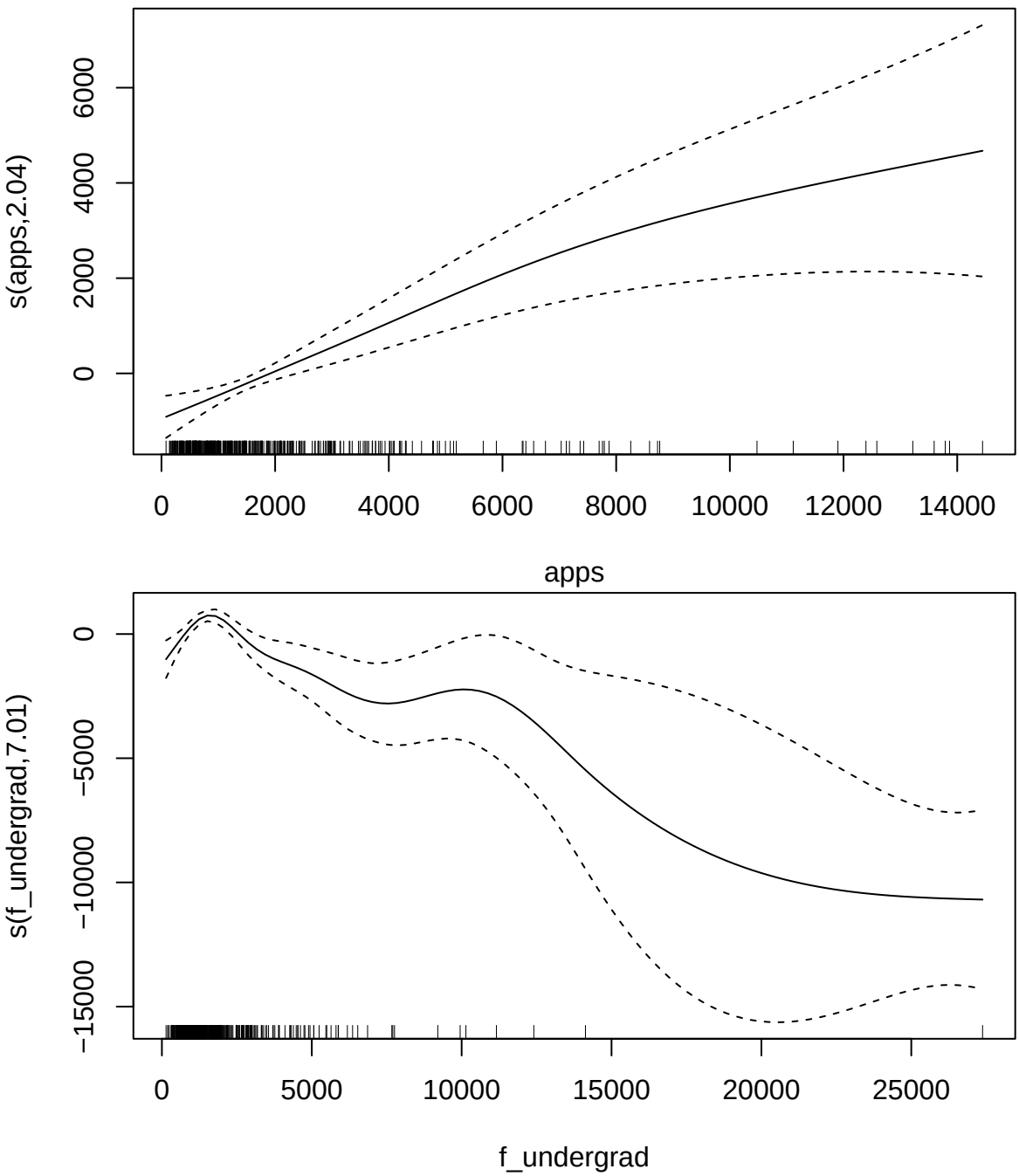
```r
gam_mars_predictors = gam(outstate ~ s(apps) + s(f_undergrad) + s(room_board) +
                            s(perc_alumni) + s(expend),
                          data = college_train)

gam_mars_predictors_interaction = gam(outstate ~ te(apps, f_undergrad) +
                                        te(room_board, expend) + s(perc_alumni),
                                      data = college_train)

anova_test = anova(gam_mars_predictors, gam_mars_predictors_interaction,
                   test = "F")
anova_test
```

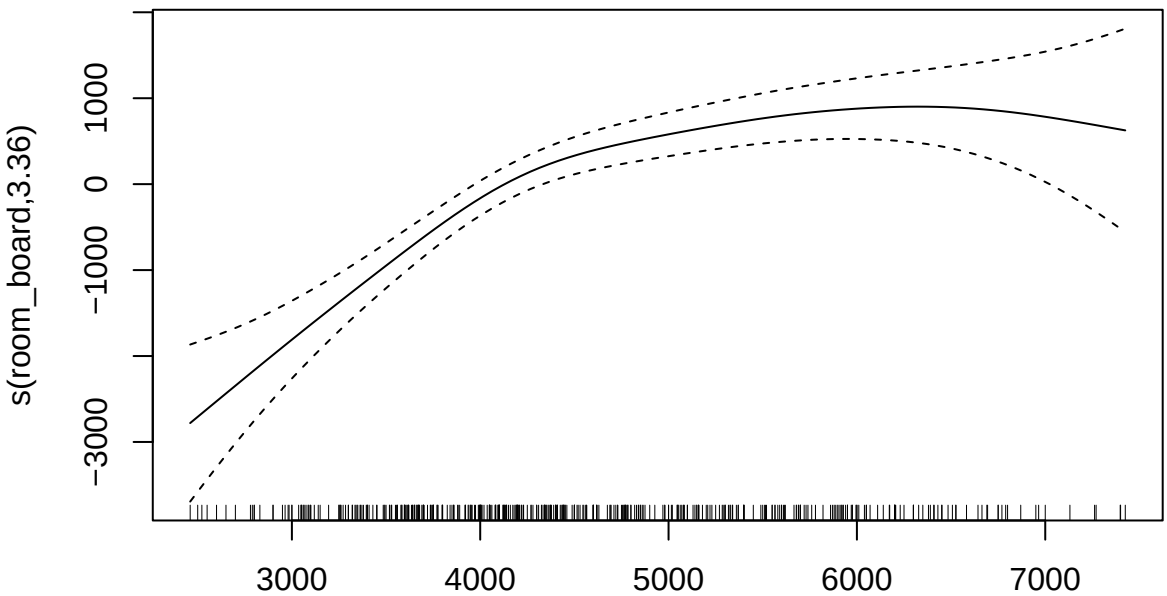
```
## Analysis of Deviance Table
##
## Model 1: outstate ~ s(apps) + s(f_undergrad) + s(room_board) + s(perc_alumni) +
##     s(expend)
## Model 2: outstate ~ te(apps, f_undergrad) + te(room_board, expend) + s(perc_alumni)
##   Resid. Df Resid. Dev     Df Deviance      F Pr(>F)
## 1    426.06 1252057915
## 2    421.74 1229732807 4.3183 22325108 1.7889 0.1247
```

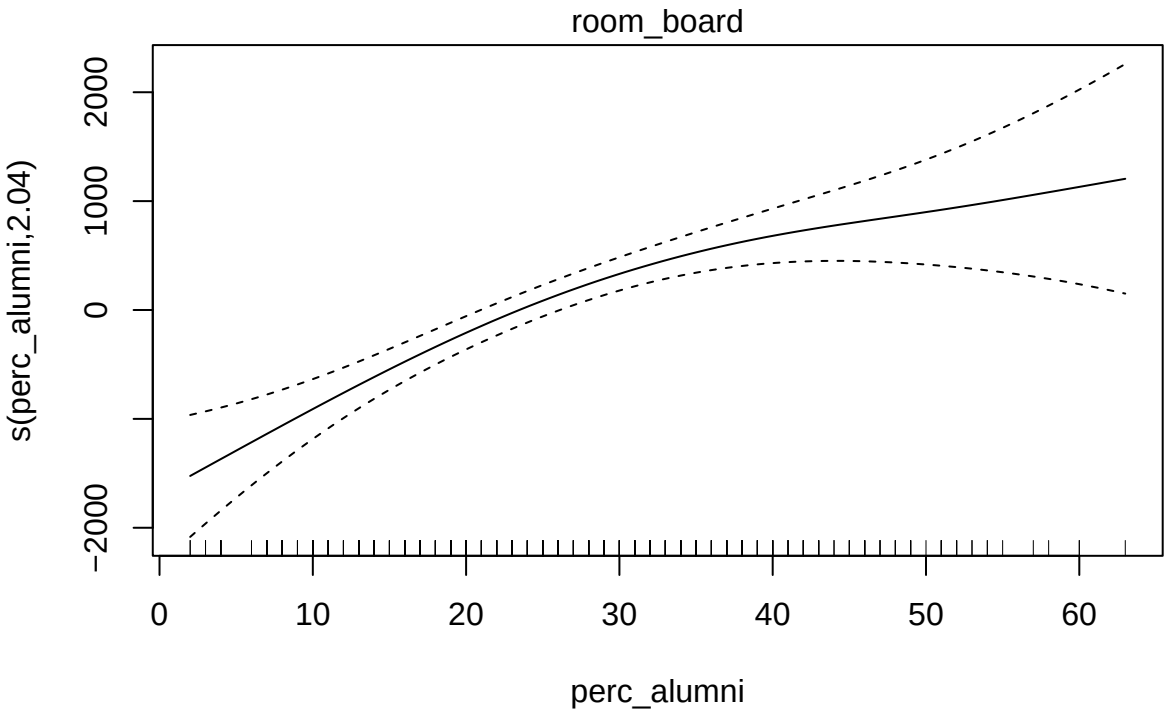
```r
p_val = anova_test$`Pr(>F)`[2]
```

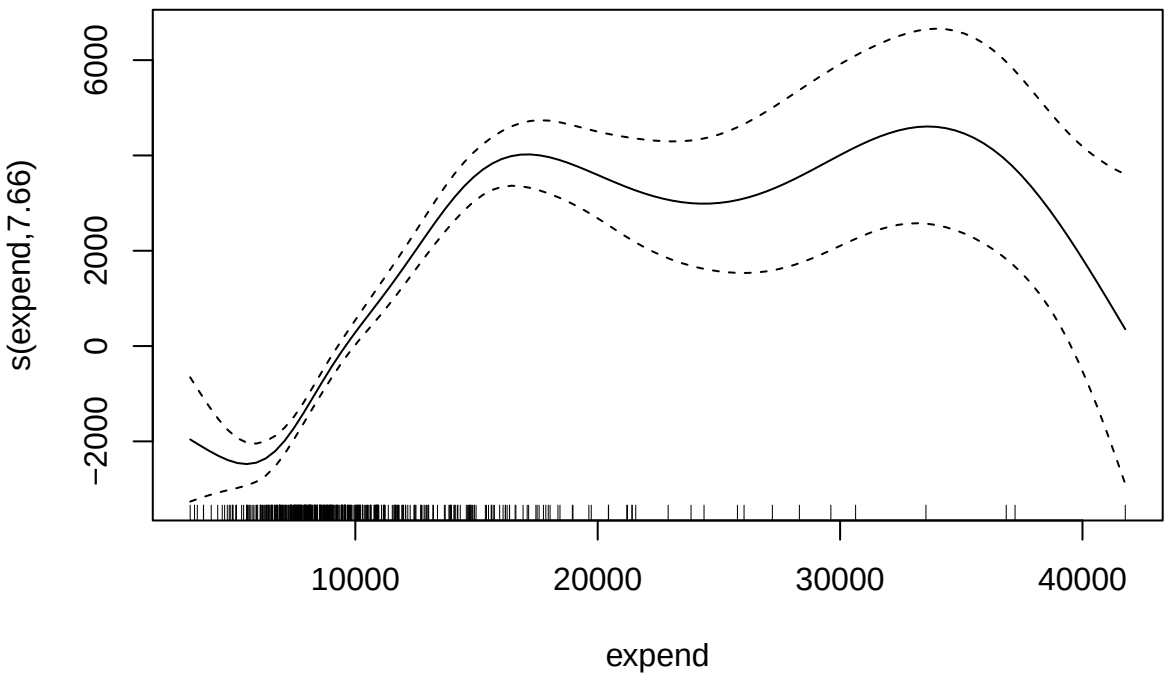
p = 0.12, which is larger than 0.05. This indicates that adding the complexity with interactions does not improve the fit to the training data.

```r
plot(gam_mars_predictors, scale = 0)
```


s(apps,2.04)
0
2000
4000
6000
0
2000
4000
6000
8000
10000
12000
14000
apps
s(f_undergrad,7.01)
0
−5000
−10000
−15000
0
5000
10000
15000
20000
25000
f_undergrad

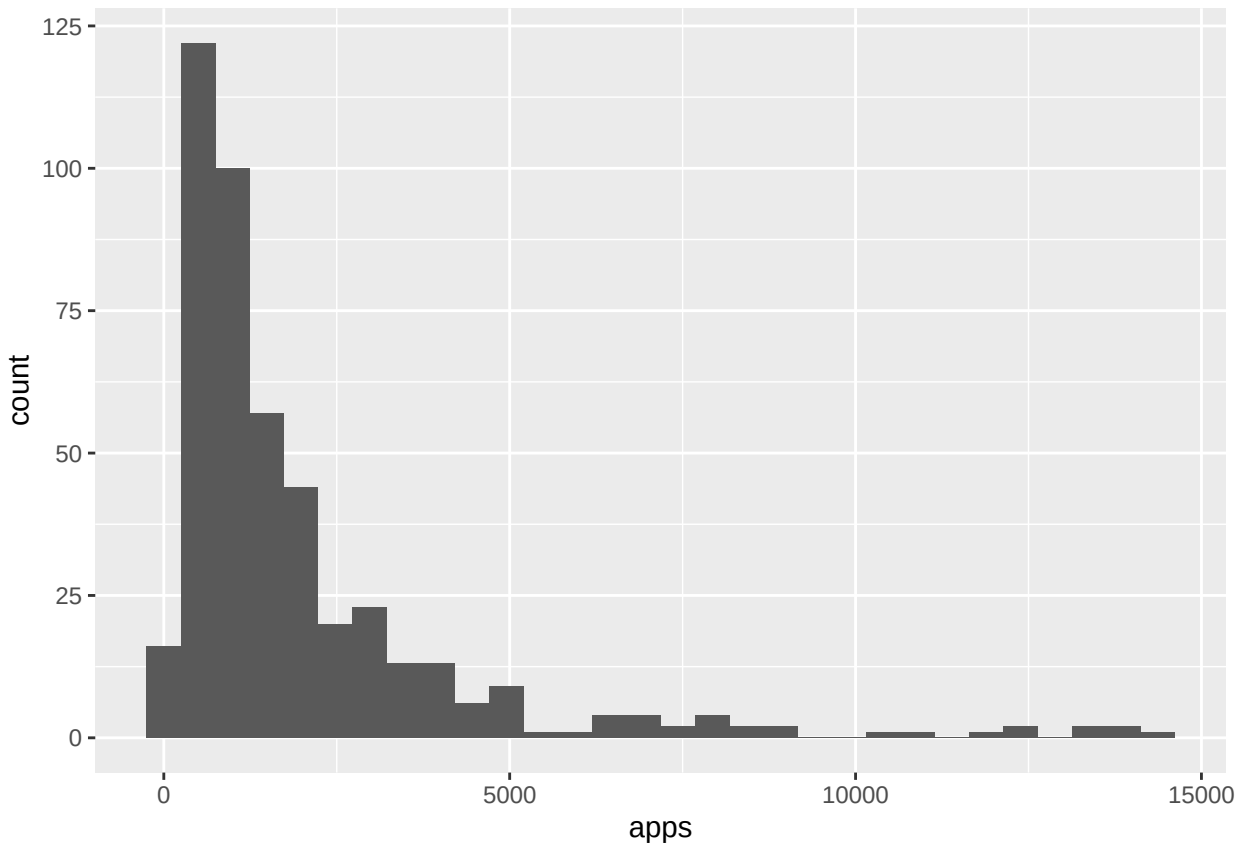

s(room_board,3.36)
1000
0
−1000
−3000
3000
4000
5000
6000
7000
room_board


s(perc_alumni,2.04)
2000
1000
0
−2000
0
10
20
30
40
50
60
perc_alumni



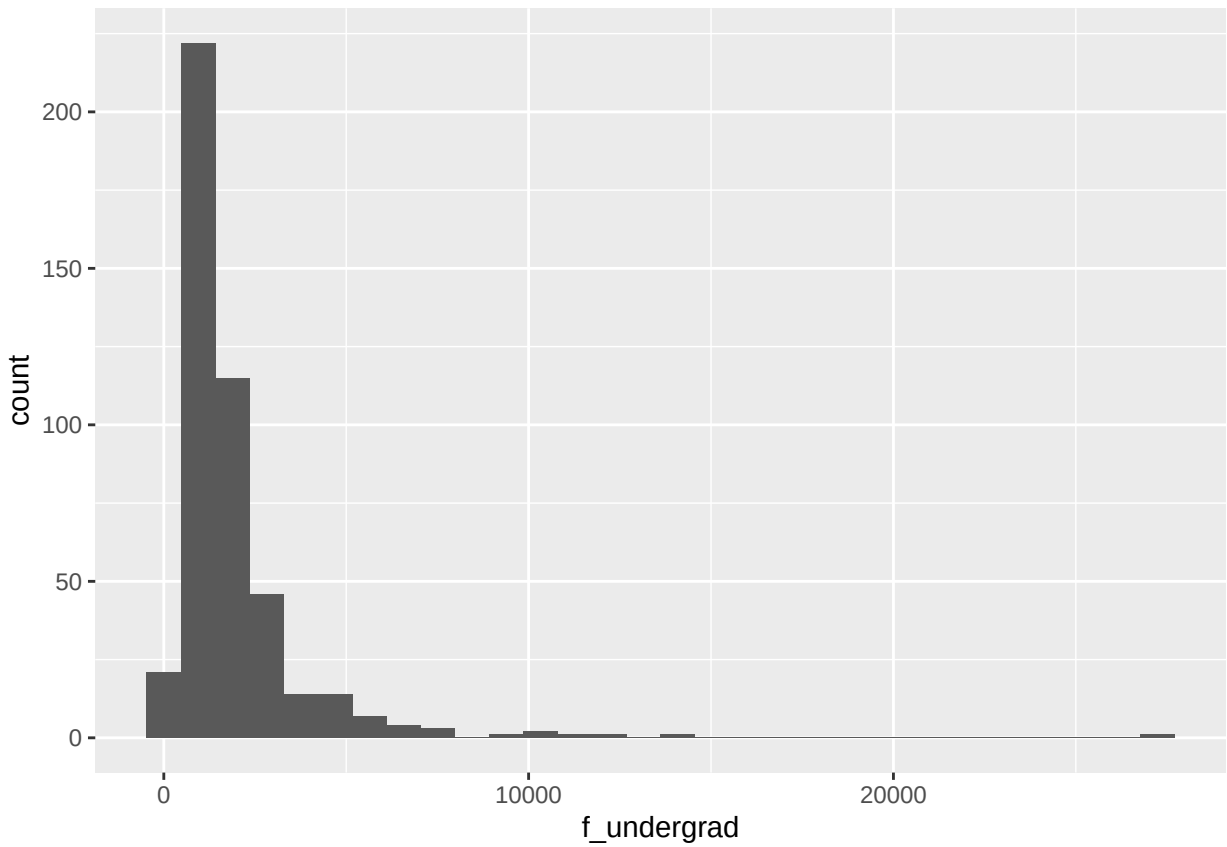

```
n_colleges = nrow(college_train)

college_train |>
  ggplot(aes(x = apps)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

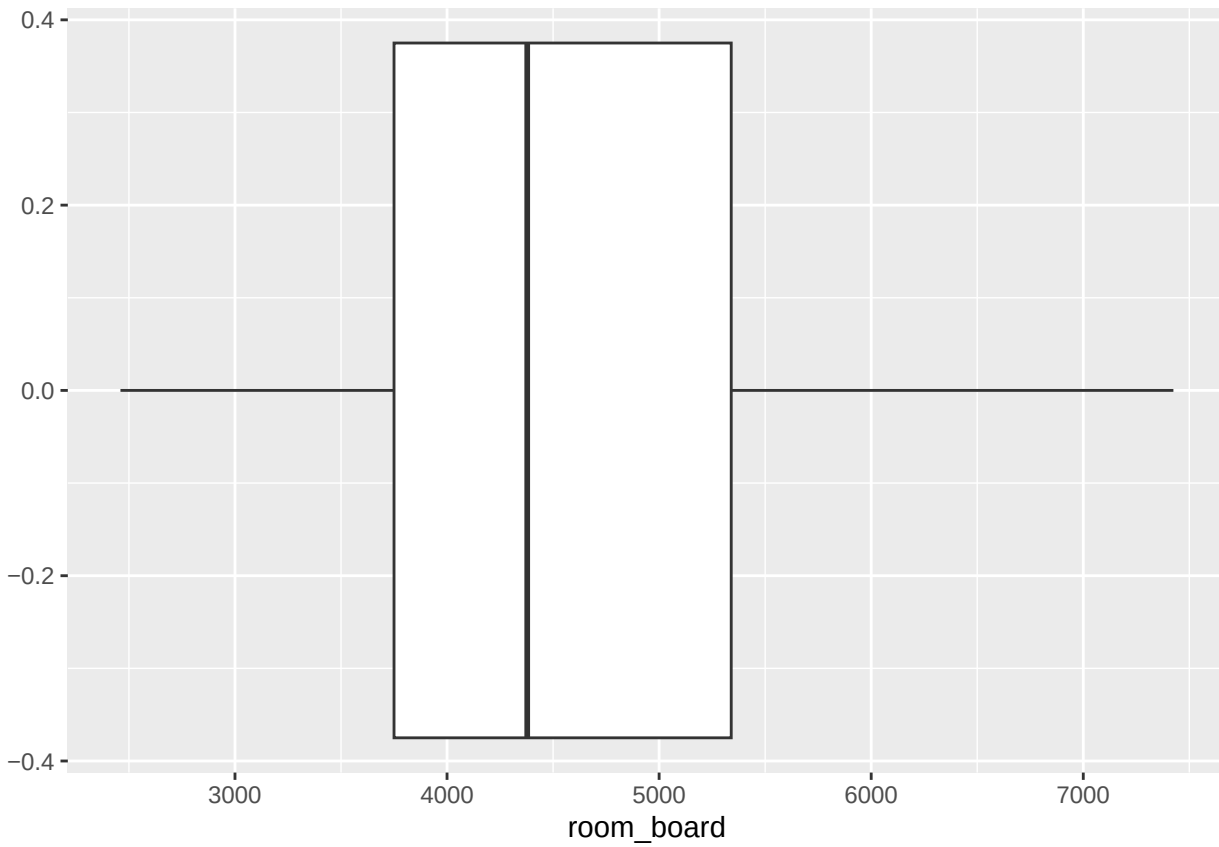



```
n_high_apps = college_train |> filter(apps > 10000) |> count() |> pull(n)

college_train |>
  ggplot(aes(x = f_undergrad)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

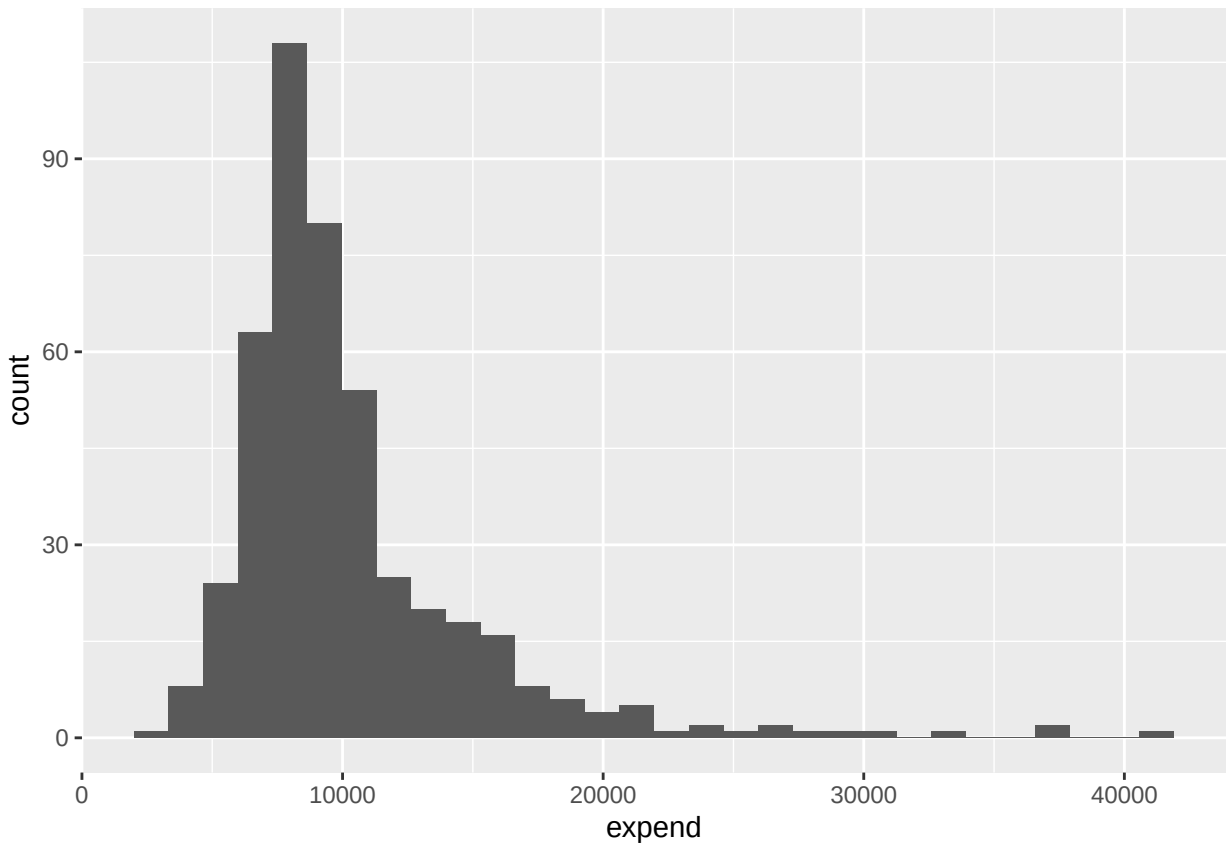



```
n_large_school = college_train |> filter(f_undergrad > 10000) |> count() |> pull(n)

college_train |>
  ggplot(aes(x = room_board)) +
  geom_boxplot()
```




```
college_train |>
  ggplot(aes(x = expend)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
n_high_expend = college_train |> filter(expend > 17000) |> count() |> pull(n)
```

Notes on plots:

- `apps`: Out-of-state tuition increases steadily as the number of applications received increases. This trend is slightly non-linear and there is a lot of variation as the number of applications received increases. This makes sense given that only 2.21% colleges have more than 10,000 applications received.
- `f_undergrad`: Out-of-state tuition drops significantly as the number of full-time undergraduates increases. This trend is highly non-linear and there is a lot of variation as the number of full-time undergraduates increases. This makes sense given that only 1.1% colleges have more than 10,000 full-time undergraduate students.
- `room_board`: Out-of-state tuition increases as costs for room and board increase, but then tapers out ~$5300, which is near the upper quartile of room and board costs
- `perc_alumni`: Tuition appears to have a fairly linear, positive relationship with the percentage of alumni who donate.
- `expend`: Out-of-state tuition increases significantly with higher instructional expenditure per student. This trend becomes more non-linear toward the higher expenditures. This makes sense given that only 7.73% colleges have more than $17,000 instructional expenditure per student.

```
gam_preds = predict(gam_mars_predictors, newdata = college_test)
gam_rmse = RMSE(gam_preds, college_test$outstate)
```

The RMSE for the selected GAM model is $2404.36.

D - MARS vs. Linear - Example

In this data set, a MARS model makes more sense. The GAM model showed that there are non-linear relationships between OOS tuition and full-time undergraduate students, instructional expenditure, and applications received. The MARS model captures these non-linear relationships. A lot of these predictors are also right skewed, with few observations at the higher end points. In a linear model, these points would have high leverage over the slopes calculated, therefore affecting the final coefficients estimated. Because the MARS model utilizes hinge functions at specific knots, the leverage from outlier points is minimal. Finally, the MARS model penalizes a more complex model, rather than forcing all the predictors into the model. This makes the model more parsimonious.

E - MARS vs. Linear - Reality

While the MARS model is more flexible and can capture non-linear relationships, a linear model is usually more practical. It is very easy to interpret and communicate the results of a linear model (e.g. "A unit increase in X predictor leads to a β increase in Y outcome variable"). It is a bit more complicated to communicate the practical translation of hinge functions and knots. It is also generally easier to avoid overfitting a linear model by employing penalized regression.